

# Gesture Controlled Presentation Slider with Voice Security

Harsha

Department of Electronics and Communication Engineering  
Government Engineering College Palakkad, Kerala  
Email: harshamoideen@gmail.com

Ms. Mehak Majed

Mentor at SURE ProED, ANDHRA PRADESH  
Jr. Embedded Systems Engineer, ATFAAL Innovations

**Abstract**—This paper presents the design and implementation of a secure and touchless gesture-controlled presentation system based on the ESP32 microcontroller. The proposed system integrates a vision-based gesture sensing module along with a microphone-based clap detection mechanism to enable intuitive human-computer interaction without the need for conventional input devices such as keyboards, mice, or handheld clickers. The clap detection module acts as an authentication layer, where a predefined clap pattern is used to unlock the system, thereby preventing unintended or accidental gesture activation.

Once authenticated, the system employs an APDS-9960 gesture sensor to detect directional hand movements, including left, right, up, and down gestures. These gestures are processed using a threshold-based and comparative algorithm to determine motion direction with high reliability. The ESP32 microcontroller translates the detected gestures into control commands and transmits them wirelessly via Bluetooth Low Energy (BLE), allowing seamless integration with presentation software on a host device.

The system architecture consists of three primary layers: input acquisition, signal processing, and wireless command execution. Special emphasis is placed on real-time responsiveness, noise filtering in clap detection, and robust gesture recognition under varying environmental conditions. The implementation eliminates the need for physical contact, making it particularly useful in smart classrooms, conference environments, and assistive technologies.

Experimental evaluation demonstrates that the system achieves high gesture recognition accuracy with minimal latency, ensuring smooth slide transitions and reliable operation. The proposed solution offers a low-cost, portable, and energy-efficient alternative to traditional presentation controllers, while also introducing an additional layer of interaction security through sound-based activation. Future enhancements may include multi-gesture customization, machine learning-based gesture classification, and cross-platform compatibility for broader application scenarios.

**Index Terms**—ESP32, Gesture Recognition, Bluetooth HID, Presentation Remote, APDS9960, Clap Detection

## I. INTRODUCTION

Human-computer interaction (HCI) has undergone significant transformation with the emergence of natural user interfaces, particularly those based on gestures and voice. In conventional presentation environments, input devices such as keyboards, mice, and handheld remotes impose physical constraints on user movement and often limit audience engagement. These traditional interfaces require direct interaction,

which can disrupt the natural flow of presentation delivery and reduce overall effectiveness. [9]

Gesture-based interaction systems provide a more intuitive and seamless alternative by enabling users to communicate with digital systems through natural hand movements. Such systems eliminate the dependency on physical input devices and enhance user mobility, making them particularly suitable for dynamic environments such as classrooms, conferences, and public speaking scenarios.

In this work, a compact and wireless gesture-controlled presentation system based on the ESP32 microcontroller is proposed. The system integrates a gesture sensing module with a microphone-based clap detection mechanism to enable touchless and secure control of presentation slides. The clap detection feature acts as an activation or authentication layer, ensuring that gesture commands are only recognized after a valid clap sequence is detected, thereby reducing unintended operations.

The ESP32 processes gesture inputs and transmits corresponding control commands via Bluetooth Low Energy (BLE) to a host device, allowing seamless slide navigation. By combining gesture recognition with sound-based activation, the proposed system enhances flexibility, improves user interaction, and provides a low-cost, portable alternative to traditional presentation controllers.

## II. LITERATURE REVIEW

Existing gesture recognition systems predominantly rely on camera-based vision techniques, which involve image processing and computer vision algorithms. While such approaches offer high flexibility and rich gesture interpretation, they require significant computational resources and are highly sensitive to environmental factors such as lighting conditions, background complexity, and occlusions. These limitations make them less suitable for low-power embedded systems and real-time portable applications. [8]

Alternatively, wearable-based gesture systems have been explored, where sensors such as accelerometers and gyroscopes are attached to the user's hand or body. Although these systems can provide accurate motion tracking, they introduce usability challenges by requiring users to wear additional hardware, which may reduce comfort and limit practical adoption in everyday scenarios.

Sensor-based approaches using dedicated gesture sensors, such as the APDS-9960, present a cost-effective and efficient solution for embedded applications. These sensors utilize infrared proximity and directional detection techniques to recognize hand movements with minimal processing overhead, making them ideal for microcontroller-based systems. Prior research demonstrates that integrating such sensors with microcontrollers enables reliable real-time gesture recognition with low latency and reduced power consumption. [6]

However, many existing implementations are limited to basic gesture detection and lack seamless integration with wireless human interface protocols. In particular, the use of Bluetooth Low Energy (BLE) Human Interface Device (HID) functionality for direct presentation control remains underexplored. This work addresses this gap by combining gesture recognition with BLE-based control, enabling a fully wireless, portable, and user-friendly presentation system with enhanced interaction capabilities.

### III. SYSTEM ARCHITECTURE

The overall system architecture is shown in Fig. 5.

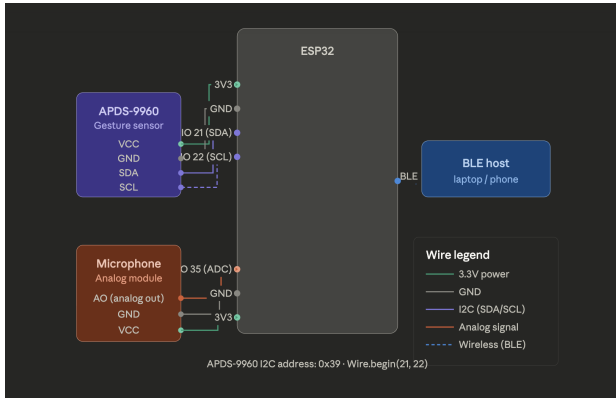


Fig. 1. System Block Diagram

The system consists of four main modules:

- Input Module (Gesture + Clap sensors)
- Processing Unit (ESP32)
- Communication Module (Bluetooth HID)
- Output Device (Laptop/PC)

### IV. HARDWARE SETUP

The hardware setup includes:

#### A. ESP32

The ESP32 is a low-cost, highly integrated microcontroller developed by Espressif Systems, featuring built-in Wi-Fi and Bluetooth connectivity. It operates at a nominal voltage of 3.3V and is powered by a dual-core 32-bit Xtensa LX6 processor, enabling efficient execution of real-time embedded applications. The microcontroller supports a wide range of peripheral interfaces, including General Purpose Input/Output (GPIO), Analog-to-Digital Converter (ADC),



Fig. 2. ESP32

Digital-to-Analog Converter (DAC), SPI, I<sup>2</sup>C, UART, and PWM.

In addition to its processing capabilities, the ESP32 provides integrated Bluetooth Low Energy (BLE) support, making it suitable for wireless communication applications such as human interface devices (HID). Its low power consumption modes and high processing performance make it an ideal choice for IoT-based systems and portable embedded solutions. Furthermore, the availability of development frameworks such as Arduino IDE and PlatformIO simplifies firmware development and deployment.

Due to its versatility, cost-effectiveness, and rich feature set, the ESP32 is widely used in applications including home automation, wearable devices, sensor networks, and wireless control systems.. [1]

#### B. APDS9960 Gesture Sensor

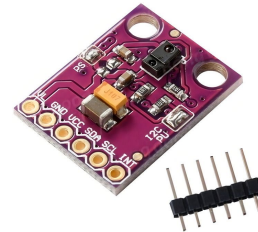


Fig. 3. APDS9960 Gesture Sensor

The APDS-9960 (GY-9960) is an integrated gesture, proximity, ambient light, and RGB color sensor that enables reliable gesture detection using infrared (IR) sensing technology. It is capable of detecting directional hand movements such as up, down, left, and right by analyzing variations in reflected infrared light intensity across multiple photodiodes.

The sensor operates by emitting infrared light through an onboard LED and measuring the reflected signals received from different directions. Based on the relative intensity of these signals, the sensor determines the direction of motion with low latency and minimal computational overhead.

Communication between the APDS-9960 sensor and the ESP32 microcontroller is achieved using the Inter-Integrated Circuit (I<sup>2</sup>C) protocol, which enables efficient data transfer with minimal wiring requirements. The sensor typically operates at a supply voltage of 3.3V, making it directly compatible with the ESP32.

Due to its compact design, low power consumption, and integrated gesture recognition capability, the APDS-9960 is widely used in embedded systems, touchless interfaces, and human-computer interaction applications [2].

### C. Microphone Sensor (MAX9814)



Fig. 4. Microphone Sensor (MAX9814)

The microphone module based on the MAX9814 high-performance amplifier is used for sound acquisition and clap pattern detection in the proposed system. The MAX9814 integrates an automatic gain control (AGC) mechanism, which dynamically adjusts the amplification of the input audio signal to maintain a consistent output level under varying ambient noise conditions.

The module captures acoustic signals through an electret microphone and converts them into an analog voltage proportional to the sound intensity. This analog output is interfaced with one of the ESP32's analog-to-digital converter (ADC) pins, enabling continuous monitoring of sound levels.

Clap detection is achieved by identifying sudden peaks in the audio signal that exceed a predefined threshold. The AGC feature of the MAX9814 improves the reliability of detection by compensating for fluctuations in background noise and signal amplitude.

Due to its high sensitivity, low noise performance, and built-in signal conditioning, the MAX9814 module is well suited for real-time sound-based interaction systems such as clap recognition and voice-triggered control [10].

## V. SOFTWARE IMPLEMENTATION

The software for the proposed system is developed using the PlatformIO development environment integrated with Visual Studio Code. PlatformIO provides a structured and efficient framework for embedded system development, including library management, code compilation, and firmware uploading to the ESP32.

### A. Gesture Detection

The APDS-9960 gesture sensor provides directional data based on infrared signal variations. The ESP32 reads this data via the I<sup>2</sup>C interface and processes it to determine the direction of motion. The detected gestures are mapped to predefined actions such as slide navigation.

### B. Clap Detection Algorithm

Clap detection is implemented using a threshold-based signal processing technique. The analog input from the MAX9814 microphone module is continuously monitored, and a clap event is detected when the signal exceeds a predefined threshold:

$$V_{\text{input}} > V_{\text{threshold}} \quad (1)$$

To ensure accurate detection, multiple claps are identified based on the time intervals between successive peaks. A valid clap pattern (e.g., double or triple clap) is used to activate or deactivate the system.

### C. Bluetooth Communication

The ESP32 is configured to operate as a Bluetooth Low Energy (BLE) Human Interface Device (HID), enabling it to emulate a wireless keyboard. Gesture inputs are translated into corresponding key commands, such as arrow keys, which are transmitted to the connected host device for slide control.

BLE communication is implemented using the PlatformIO development environment, which facilitates efficient configuration and deployment of embedded applications. The use of BLE ensures low power consumption and reliable wireless communication for real-time interaction [1], [3], [5].

## VI. WORKING PRINCIPLE

The proposed system operates through a sequential process involving system initialization, authentication using clap detection, gesture recognition, and wireless command transmission. The overall workflow ensures secure and intuitive interaction for presentation control.

### A. System Initialization

At startup, the ESP32 microcontroller initializes all connected peripherals, including the APDS-9960 gesture sensor and the microphone module. The Bluetooth Low Energy (BLE) interface is configured in Human Interface Device (HID) mode, allowing the system to function as a wireless keyboard for the host device.

### B. Clap-Based Activation

To prevent unintended gesture execution, a clap detection mechanism is implemented as an activation layer. The microphone continuously monitors ambient sound levels, and a clap is identified when the signal exceeds a predefined threshold.

- A predefined sequence of claps (e.g., double or triple clap) activates the system.
- If the pattern is not matched within a specific time window, the system resets to the idle state.

This mechanism ensures that gesture control is only enabled intentionally.

### C. Gesture Detection

Once activated, the APDS-9960 sensor detects hand movements in multiple directions, including left, right, up, and down. The sensor provides directional data based on infrared signal variations, which are processed to determine the gesture.

### D. Signal Processing

The ESP32 processes the incoming gesture data using a comparative algorithm to identify the dominant direction. Based on the detected gesture, the system maps the input to predefined control actions.

### E. Bluetooth Command Transmission

After gesture recognition, the ESP32 transmits the corresponding command to the connected device via BLE HID protocol. The commands emulate keyboard inputs, enabling functions such as:

- Right gesture → Next slide
- Left gesture → Previous slide
- Near gesture → Zoom in
- Far gesture → Zoom out

### F. Workflow Summary

The complete operation of the system can be summarized as follows:

- 1) Initialize sensors and BLE interface
- 2) Continuously monitor sound for clap detection
- 3) Validate clap pattern to activate system
- 4) Detect hand gesture using APDS-9960
- 5) Process gesture direction in ESP32
- 6) Transmit corresponding command via Bluetooth

This structured workflow ensures reliable, secure, and real-time gesture-based control for presentation applications.

## VII. RESULTS AND DISCUSSION

The proposed gesture-controlled presentation system was experimentally evaluated under various operating conditions to assess its performance in terms of detection range, response time, and accuracy.

### A. Gesture Detection Performance

The APDS-9960 gesture sensor demonstrated reliable detection within an effective range of 5–15 cm. Within this range, directional gestures such as left, right, up, and down were consistently recognized with minimal misclassification. Detection performance decreased beyond 15 cm due to reduced infrared signal strength and sensitivity limitations of the sensor.

### B. System Response Time

The overall system response time, measured from gesture execution to command transmission via Bluetooth, was observed to be less than 100 ms. This low latency ensures smooth and real-time interaction, which is critical for applications such as presentation control.

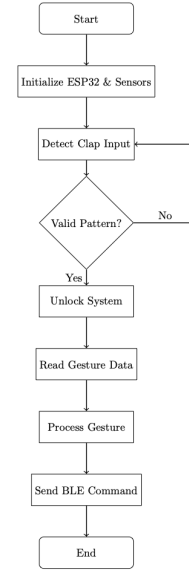


Fig. 5. System Block Diagram

### C. Accuracy Analysis

The system achieved an average gesture recognition accuracy of approximately 90

### D. Environmental Impact

System performance was influenced by environmental conditions. Strong ambient light, particularly direct sunlight, affected the infrared sensing capability of the APDS-9960, leading to occasional detection errors. Additionally, high background noise levels impacted clap detection reliability, resulting in false triggers or missed activations.

### E. Overall Discussion

The system demonstrated stable and reliable performance within indoor environments such as classrooms and meeting rooms. The integration of clap-based activation with gesture control enhances both usability and operational security. The observed low response time and acceptable accuracy validate the effectiveness of the proposed system as a practical alternative to conventional presentation controllers.

Future improvements may include adaptive thresholding, improved noise filtering, and advanced gesture recognition algorithms to enhance robustness under varying environmental conditions.

## VIII. ADVANTAGES AND LIMITATIONS

### A. Advantages

- **Low Cost:** The system utilizes affordable components such as the ESP32 and APDS-9960 sensor, making it a cost-effective alternative to commercial presentation devices.

- **Portable Design:** The compact hardware implementation allows easy portability, enabling users to carry and use the device conveniently in various environments such as classrooms and conference halls.
- **Wireless Control:** Integration of Bluetooth Low Energy (BLE) enables seamless wireless communication with host devices, eliminating the need for physical connections.
- **Touchless Interaction:** The system allows hands-free control using gestures, improving user convenience and hygiene, especially in shared environments.
- **Enhanced Security:** The clap-based activation mechanism prevents unintended gesture execution, ensuring that the system responds only to intentional user input.
- **Easy Implementation:** The system is based on simple hardware and straightforward algorithms, making it easy to design, implement, and replicate for academic or practical applications.

### B. Limitations

- **Limited Detection Range:** Gesture recognition is effective only within a short range (5–15 cm), restricting user flexibility compared to camera-based systems.
- **Sensitivity to Ambient Light:** Strong lighting conditions, especially direct sunlight, can interfere with the infrared-based gesture sensor and reduce accuracy.
- **Noise Interference:** High background noise may affect clap detection, leading to false activations or missed inputs.
- **Limited Gesture Set:** The system supports only basic directional gestures, which may limit functionality for more complex applications.
- **No Adaptive Learning:** The current implementation relies on threshold-based logic and does not incorporate machine learning for adaptive or personalized gesture recognition.

## IX. APPLICATIONS

The proposed gesture-controlled system has a wide range of applications across various domains due to its touchless interaction and wireless control capabilities.

- **Classroom Presentations:** The system enables instructors to navigate slides seamlessly without being restricted to a specific location, thereby improving teaching effectiveness and student engagement.
- **Business Meetings and Conferences:** Professionals can deliver presentations more dynamically by controlling slides through gestures, enhancing communication and reducing dependency on handheld devices.
- **Smart Home Control:** The gesture recognition mechanism can be extended to control smart home appliances such as lights, fans, and media systems, providing a convenient and contactless user interface.
- **Assistive Technologies:** The system can assist individuals with physical disabilities by offering an alternative

method of interaction with computers and electronic devices through simple hand gestures.

- **Public Interactive Systems:** It can be deployed in kiosks, exhibitions, and digital signage systems where touchless interaction is preferred for hygiene and ease of use.
- **Healthcare Environments:** In hospitals and laboratories, gesture-based control can reduce physical contact with devices, helping maintain sterile conditions and minimize contamination risks.

## X. FUTURE WORK

The proposed system demonstrates effective gesture-based control; however, several enhancements can be implemented to improve its performance, scalability, and usability.

- **Machine Learning-Based Gesture Recognition:** Future work can incorporate machine learning algorithms to enable adaptive and more accurate gesture classification. This would allow the system to learn user-specific gesture patterns and improve robustness under varying conditions.
- **Extended Detection Range:** Enhancing the sensing mechanism or integrating additional sensors can increase the gesture detection range beyond the current 5–15 cm, thereby improving user convenience and flexibility.
- **Mobile Application Integration:** Developing a dedicated mobile application can provide a user-friendly interface for configuration, customization of gesture mappings, and real-time system monitoring.
- **Advanced Noise Filtering:** Implementing digital signal processing techniques such as adaptive filtering and peak validation can improve clap detection accuracy by minimizing the impact of ambient noise.
- **Multi-Gesture and Custom Controls:** Expanding the system to support a wider set of gestures and user-defined commands can enhance functionality and adaptability for different applications.
- **Cross-Platform Compatibility:** Improving compatibility with multiple operating systems and devices will make the system more versatile and widely applicable.

These enhancements aim to transform the current prototype into a more intelligent, scalable, and user-centric interaction system suitable for real-world deployment.

## XI. CONCLUSION

This paper presented the design and implementation of a gesture-controlled presentation remote with voice-based (clap) security using the ESP32 microcontroller. The proposed system integrates gesture recognition and sound-based activation to enable a secure, touchless, and intuitive human–computer interaction mechanism. By eliminating the need for conventional input devices such as keyboards and handheld clickers, the system enhances user mobility and presentation efficiency.

The implementation demonstrates reliable gesture detection within a practical range, low-latency Bluetooth communication, and effective clap-based activation for preventing unintended operations. Experimental results confirm that the sys-

tem achieves satisfactory accuracy and responsiveness under typical indoor conditions, making it suitable for real-world applications such as classrooms, business presentations, and interactive environments.

Overall, the proposed solution provides a low-cost, portable, and energy-efficient alternative to existing presentation control systems. Future enhancements involving machine learning, improved sensing range, and advanced noise filtering can further enhance system performance and adaptability. The work highlights the potential of combining gesture recognition with sound-based security for developing next-generation interactive systems.

## REFERENCES

- [1] Espressif Systems, *ESP32 Series Datasheet*, Version 4.3, 2023. [Online]. Available: <https://www.espressif.com>
- [2] Broadcom Inc., *APDS-9960 Gesture Sensor Datasheet*. [Online]. Available: <https://www.broadcom.com>
- [3] Bluetooth Special Interest Group (SIG), *Bluetooth Low Energy HID Profile Specification*. [Online]. Available: <https://www.bluetooth.com>
- [4] Arduino, *ESP32 BLE Keyboard Library Documentation*. [Online]. Available: <https://www.arduino.cc>
- [5] PlatformIO, *PlatformIO IDE and Embedded Development Ecosystem*. [Online]. Available: <https://platformio.org>
- [6] SparkFun Electronics, *APDS-9960 Gesture Sensor Hookup Guide*. [Online]. Available: <https://learn.sparkfun.com>
- [7] Tutorials, *ESP32 Bluetooth Low Energy (BLE) Guide*. [Online]. Available: <https://randomnerdtutorials.com>
- [8] M. Banz and M. Shiloh, *Getting Started with Arduino*, 3rd ed. Maker Media, 2014.
- [9] K. Ashton, "That 'Internet of Things' Thing," *RFID Journal*, 2009.
- [10] Analog Devices, *MAX9814 Microphone Amplifier with Automatic Gain Control Datasheet*. [Online]. Available: <https://www.analog.com>